Hochschule Konstanz
University of Applied Sciences

Presentation of Team Projects 2026

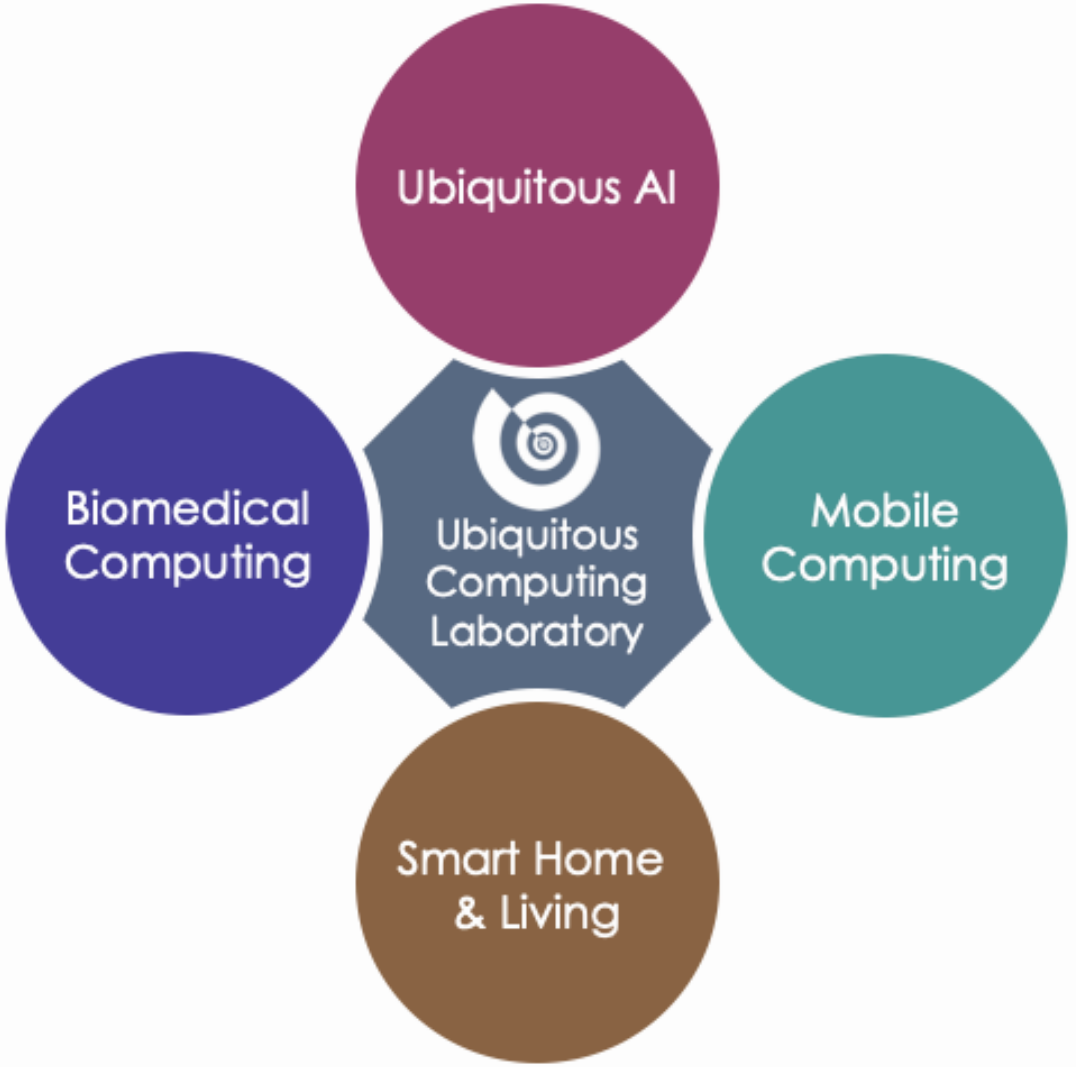


UbiquitousComputing LABORATORY

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UC Lab Overview

- Biometric sensors and sensor integration
- Biovital signal processing
- Medical IoT
- Telemedicine
- eHealth/eCare
- Body area networks
- **Artificial intelligence**
- **Explainable AI**
- **Integration to clinical practice**

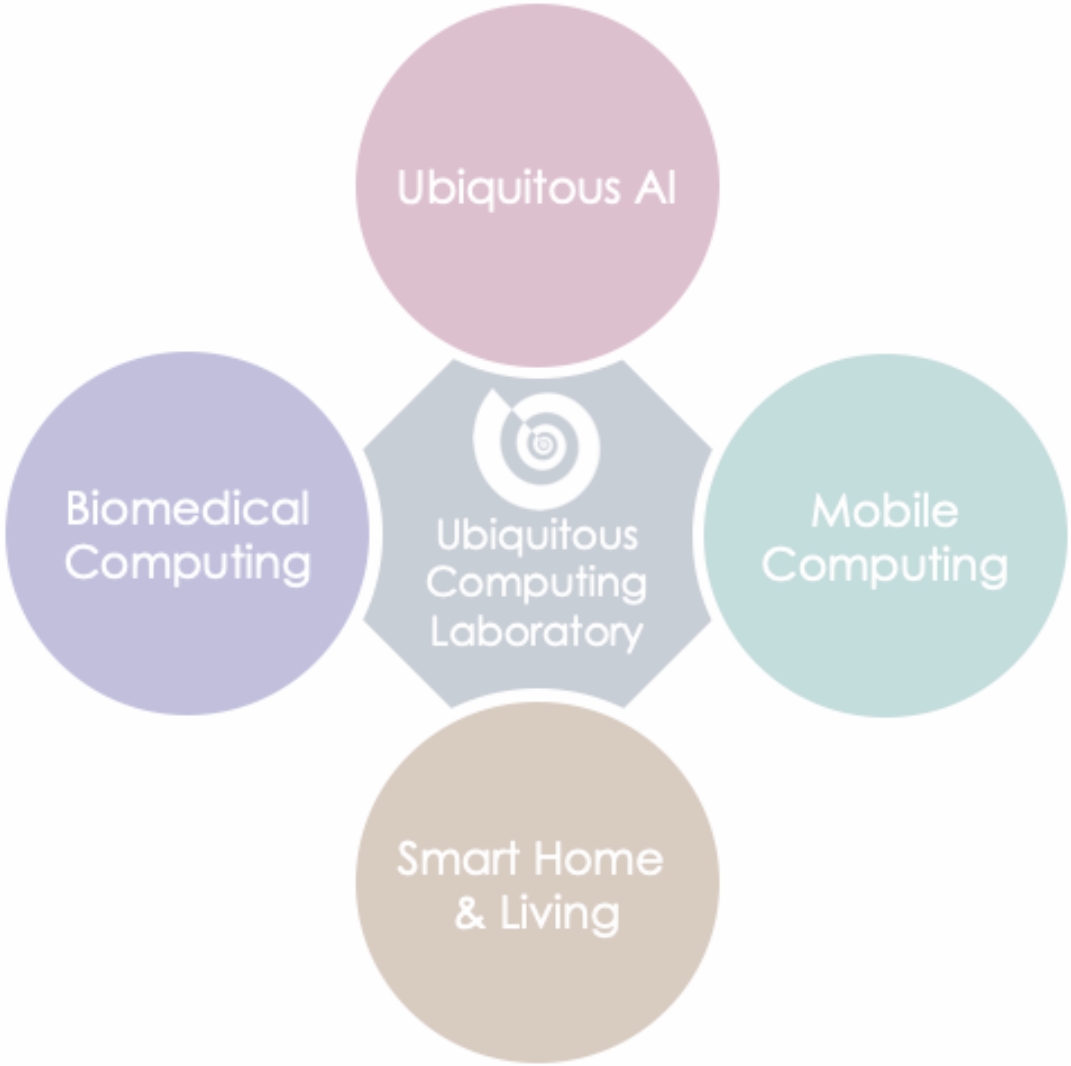


- Programming platforms for mobile devices
- Portable biometric data capturing
- Operating systems for mobile devices
- Mobile and wireless prototyping platforms
- Security for mobile devices

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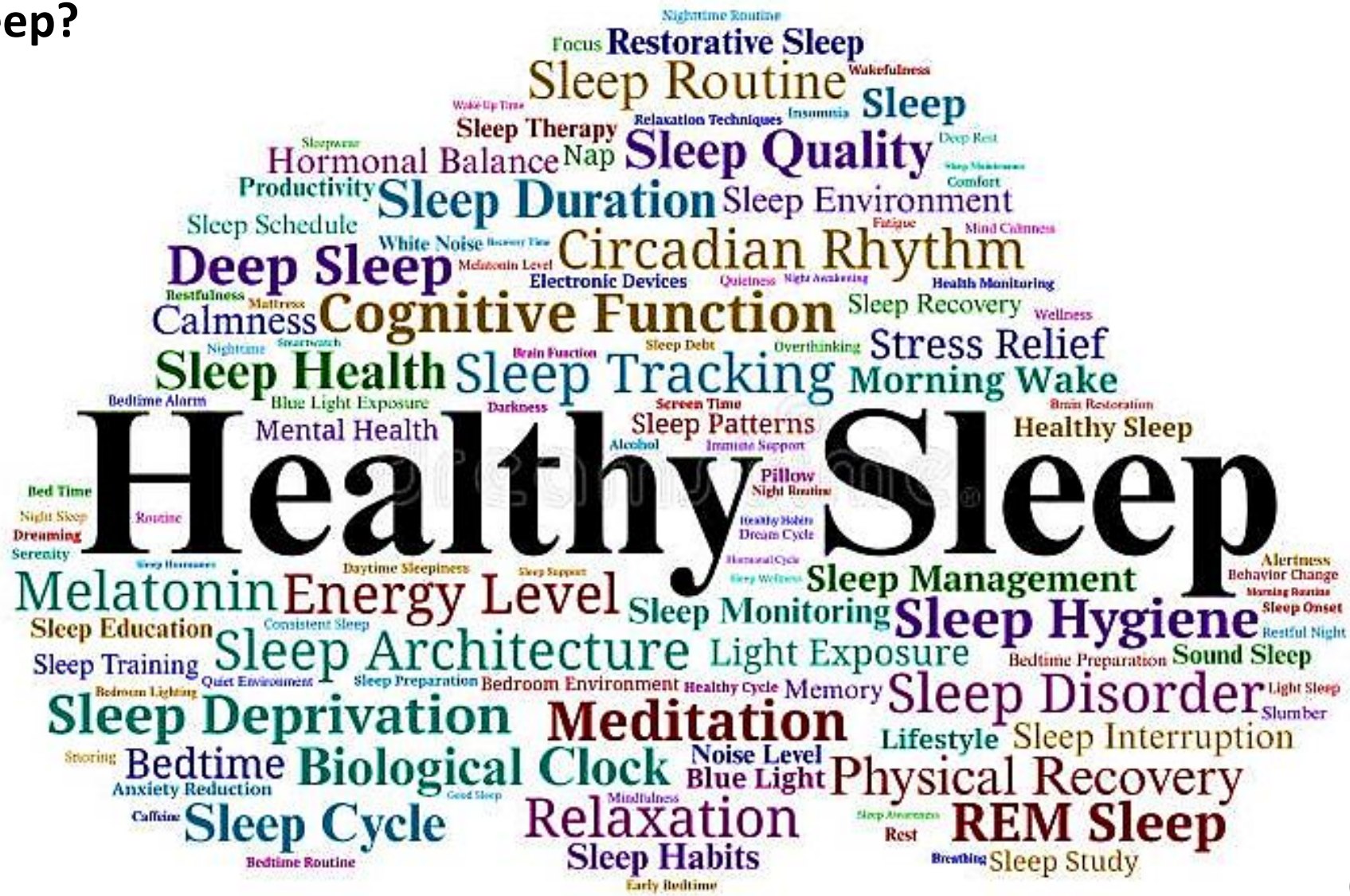


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AIN Team project

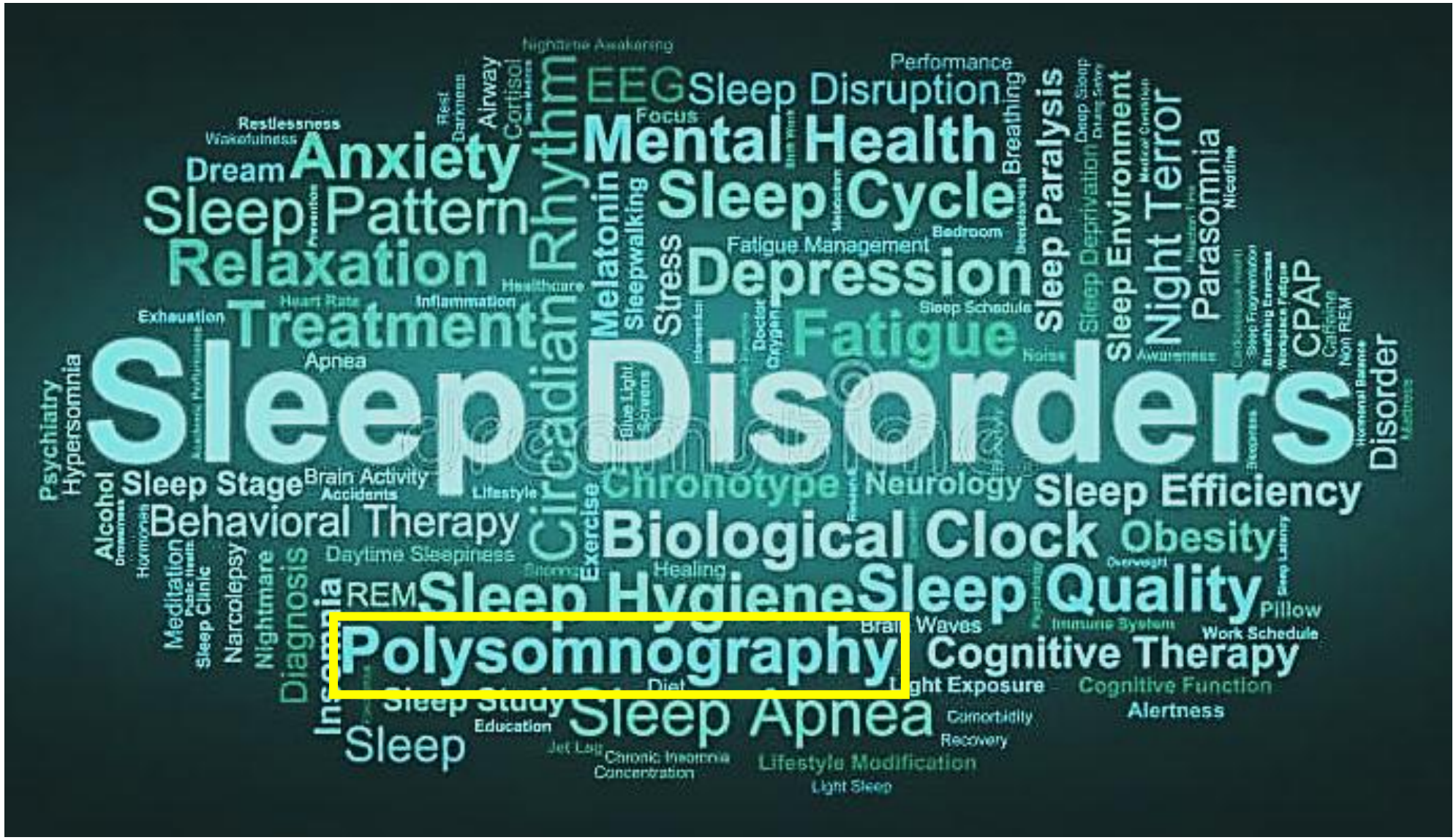
How can we leverage multimodal data to boost sleep disorder classification using machine learning?

What is sleep?



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Sleep disorders



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Polysomnography (PSG)

Gold standard for sleep assessment
POLYSOMNOGRAPHY (PSG)



Sleep clinic



Continous overnight evaluation



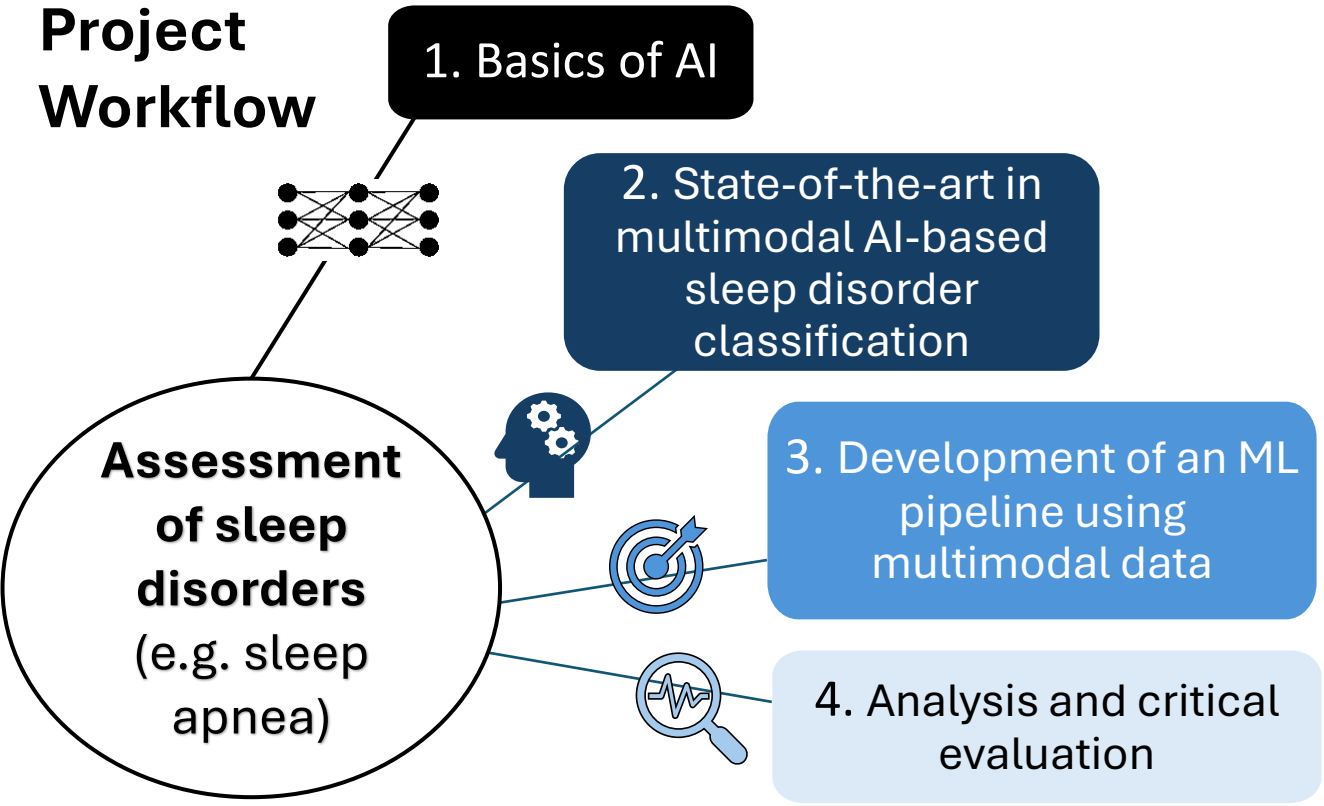
Multimodal assessment

- Electrical brain activity **(EEG)**
- Eye movement **(EOG)**
- Cardio-respiratory activity **(ECG, SpO2, Breathing)**
- Body movement **(Pressure, acceleration)**

How can we leverage multimodal data to boost sleep disorder classification using machine learning?

Goal: To develop a multimodal pipeline for the assessment of sleep disorders (e.g. sleep apnea) using a machine learning (ML) pipeline.

Project Workflow



Expected outcome



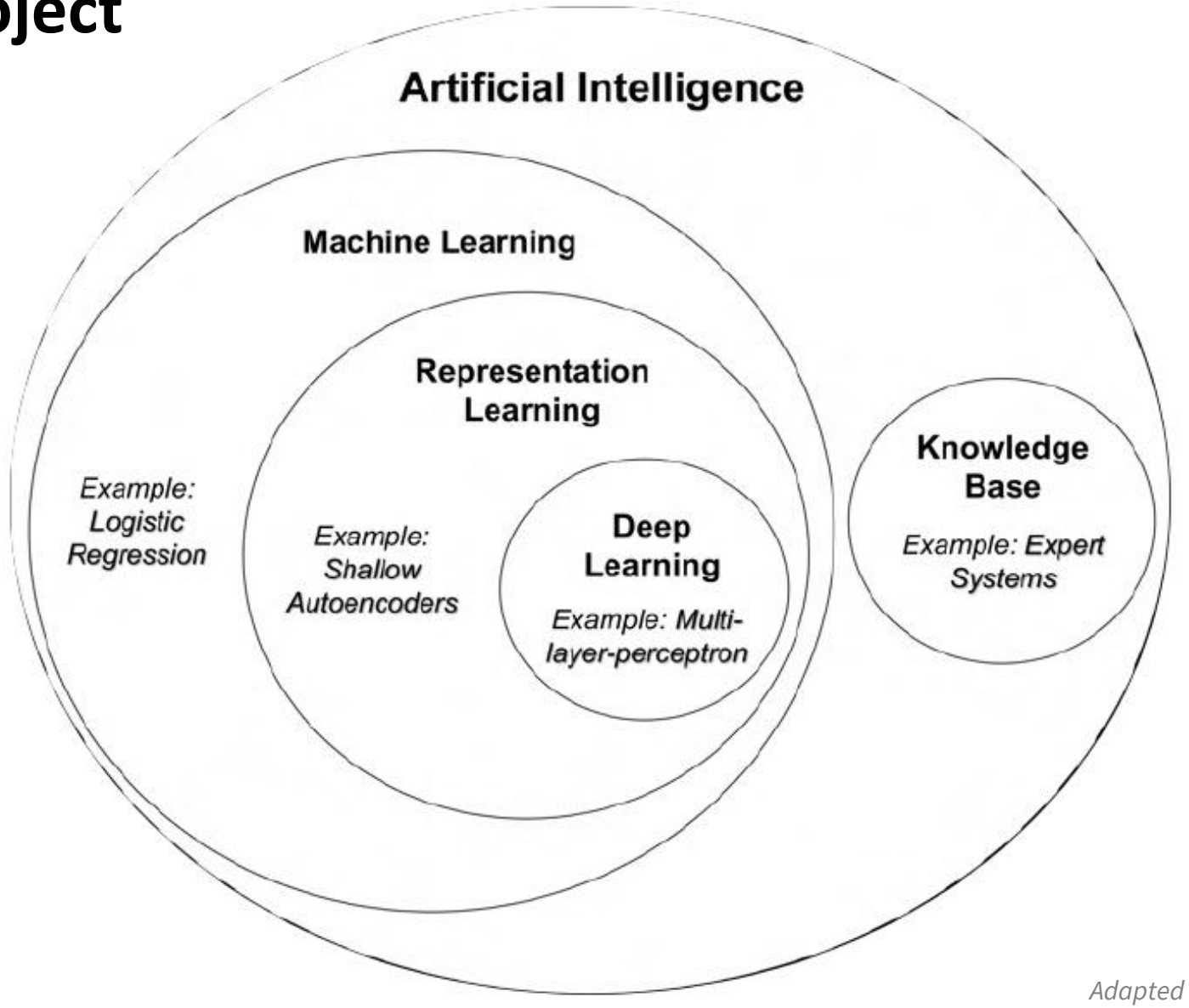
Requirements

Analytic skills, Programming skills

Focus of the project

1. Basics of AI

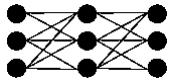
Assessment of sleep disorders



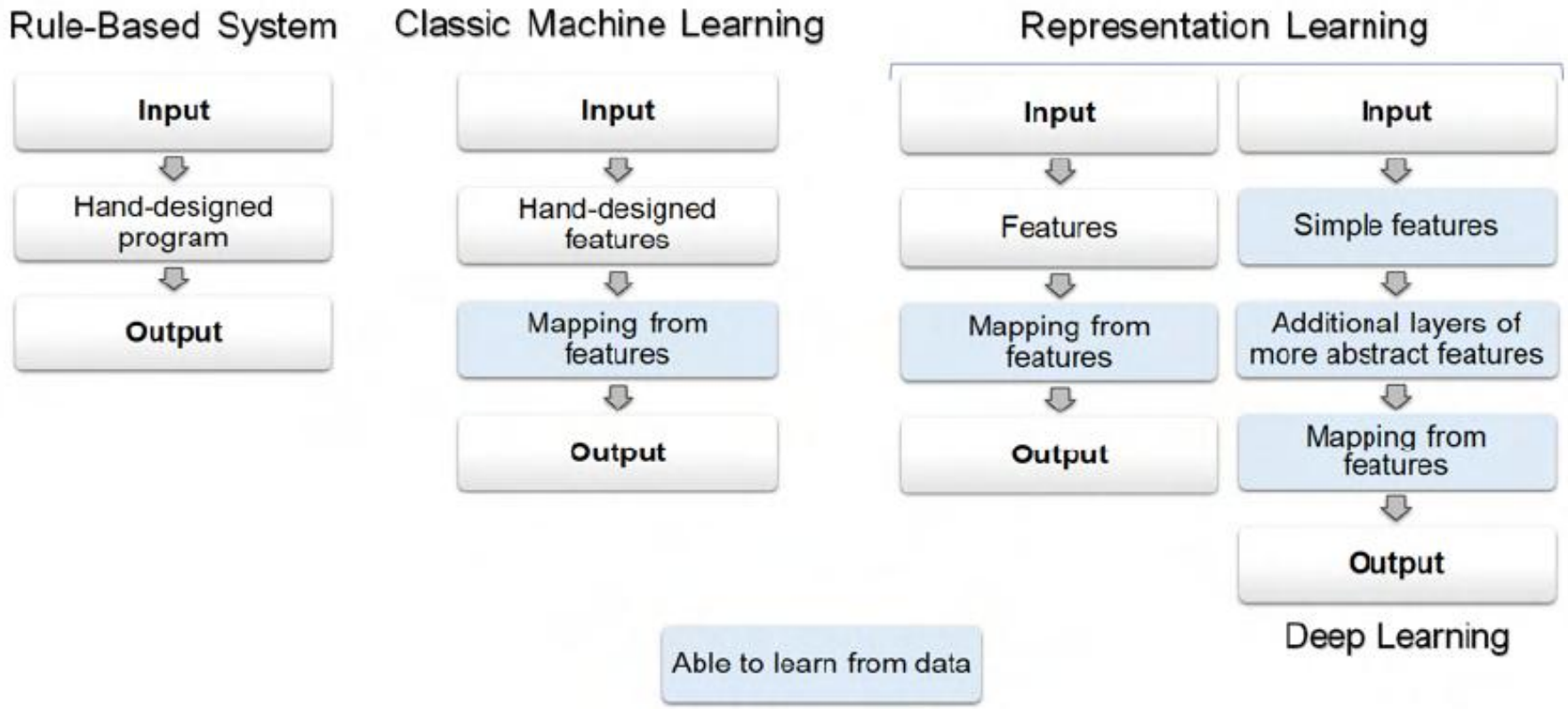
Adapted from Goodfellow et al. (2016)

Focus of the project

1. Basics of AI

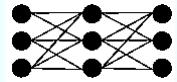


Assessment of sleep disorders



Adapted from Goodfellow et al. (2016)

Focus of the project



1. Basics of AI

- Fundamental principles and techniques



2. State-of-the-art in multimodal AI-based sleep disorder classification

- What is multimodality?
- What are the most common sleep disorders?
- What is sleep apnea? Which signals can be used to detect it?
- How can we use AI to analyze multimodal data and provide a classification of sleep apnea?

Assessment of sleep disorders



3. Development of an ML pipeline using multimodal data

- Use of public datasets available
- Leverage existing frameworks



4. Analysis and critical evaluation

- AI Performance evaluation
- Comparison with current status in the literature
- Strengths and limitations

Guidelines & Expected outcome

Example of Project structure

1. Identify key aspects of a **typical AI pipeline**
2. Acquire **basic knowledge** on sleep apnea detection and multimodality
3. **Review the** current state-of-the-art for sleep apnea detection with multimodal data and AI
4. **Develop your AI pipeline to perform sleep apnea detection with multimodal data**
5. **Analyze** your results

Guidelines & Expected outcome

Example of Project structure

1. Identify key aspects of a **typical AI pipeline**
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5. **Analyze** your results

Expected outcome

Development of an AI pipeline applicable to multimodal signals for sleep apnea classiication

- Programming In Python
- Using public available datasets



State-of-the-art → ML framework → Study positioning

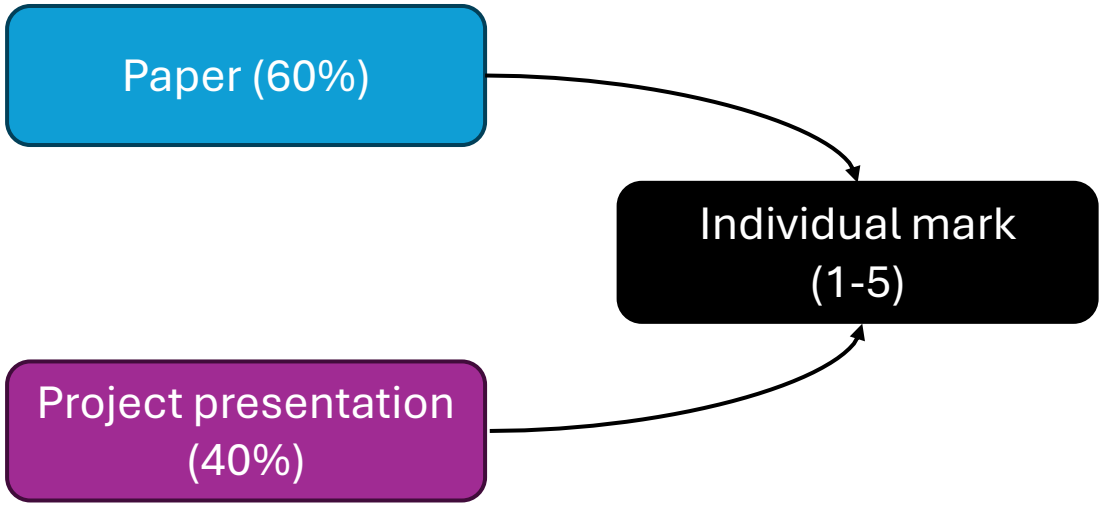
Team Project Evaluation

Each team member will receive an individual mark as described below

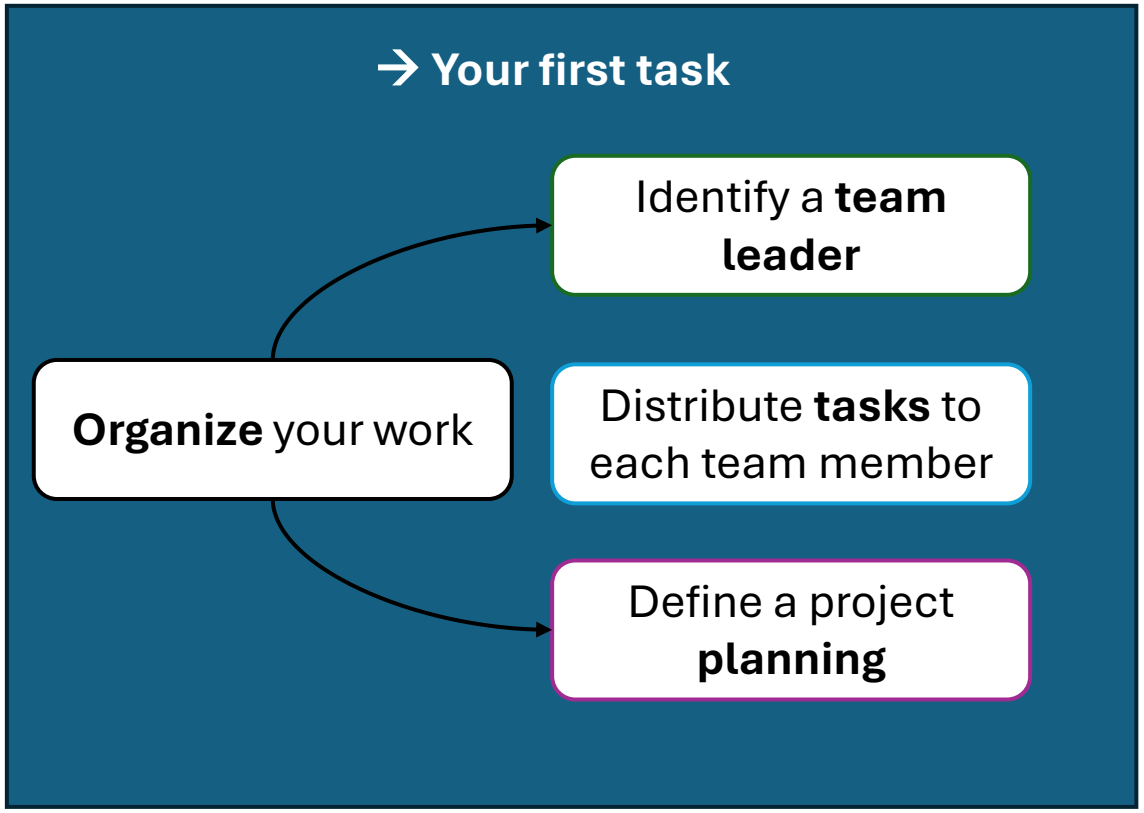
One paper per team

- Format: KES conference paper
  - Up to 10 pages (max 7-8000 words)
  - Max 5 graphical items (tables and/or figures)

- Duration: 15 min each team member

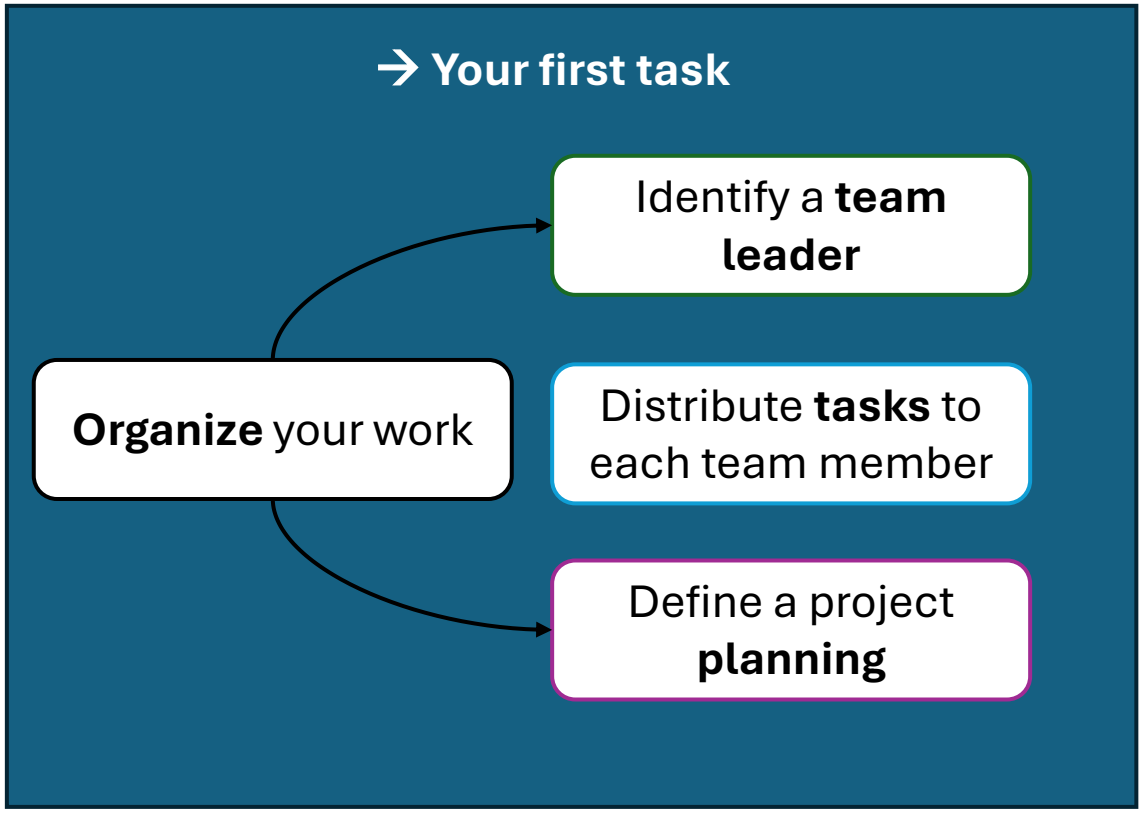


What comes next?



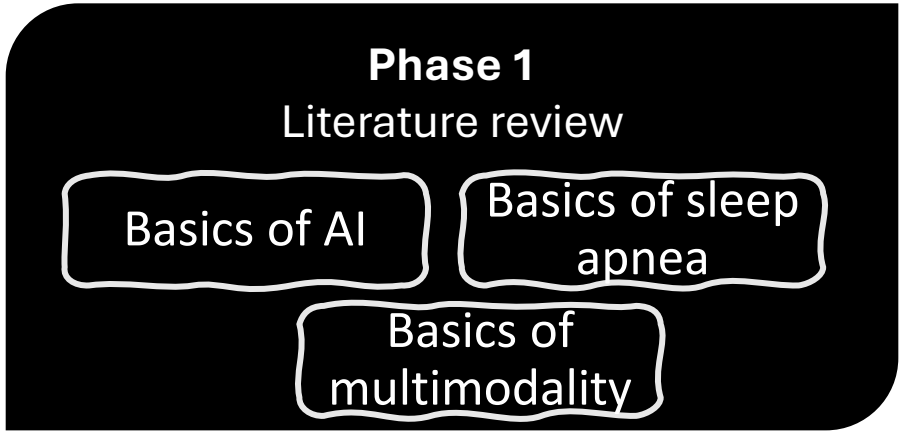
→ **Decide on a regular meeting schedule**

What comes next?



→ **Decide on a regular meeting schedule**

→ **Your project starts**



References

AI Basics and application in healthcare

- Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016. http://www.deeplearningbook.org.
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Basics of sleep

- A. Physiological basis of sleep | ESRS

Sleep apnea detection and multimodality

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